MILITARY SPECIFICATION

INDICATOR, PRESSURE, TYPE MO-2

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hermetically sealed, remote indicating, voltage ratio type, oil pressure indicator, designated Type MO-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-I-25437	Indicators, Aircraft, Pressure, Voltage Ratio Type, General Specification For
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby Synthetic
MIL-T-38230	Transmitter, Pressure TRU-66/A, Oil, Variable Reluctance

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-20T04:56Z
Provided for private study and research; the distribution of this document is unlimited.
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Requirements For Reliability Program (For Systems And Equipment)
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements.

MS STANDARDS

MS33585	Pointer, Dial, Standard Design Of Aircraft Instrument
MS33639	Cases, Instrument, Clamp Mounted, Aircraft

(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadephia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 202610-20T04:56Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for the opening of bids (See 4.3 and 6.3).

3.2 General specification. The requirements of MIL-I-25437 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction

3.3.1 Indicator. The indicator shall be designed for use with a TRU-66/A transmitter conforming to MIL-T-38230.

3.3.2 Case. The case dimensions shall conform to MS33639 for the 2-inch size. uhe case length shall be 2-625 ±0.375 inches.

3.3.3 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.3.3.1 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of 2,000 hours at a confidence level of 90 percent and a discrimination ratio of 1.5.

3.3.3.2 Longevity. The indicator shall have an opening life span of 3,000 hours.

3.4 Performance.

3.4.1 Tolerance. The indicator shall operate within the tolerance specified in tables I and II.

3.5 Pointer. The pointer shall be in accordance with MS33585-8 except that the length shall be such that the tip will extend into the shortest graduation a distance approximately one-third the length of that graduation.

3.5.1 Pointer finish. The shaded portion of the pointer shall be finished in white, color No. 37875 of FED-STD-595.

3.6 Dial. The dial shall conform to figure 1. A minimum diameter of 1-750 inches shall be maintained across the outside of the graduations. The graduations for 45 psi shall be located on the horizontal centerline at the left side of the dial.

Source: https://assist.dla.mil -- Downloaded: 202610-20T04:56Z
Check the source to verify that this is the current version before use.

TABLE I. Temperature Scale Error.

Test Pressure PSI	Scale Error	
	Room Temperature PSI	High and Low Temp Exposure High and Low Temperature PSI
0	1.5	3.0
10	1.5	3.0
20	1.5	3.0
30	1.5	3.0
40	1.5	3.0
50	1.5	3.0
60	1.5	3.0
70	1.5	3.0
80	1.5	3.0
90	1.5	3.0
100	1.5	3.0

TABLE II. Position, Friction, and Vibration Error.

Test	Tolerance ± PSI
Position error at 50 psi	1.5
Friction	
Room temperature	1.5
High temperature	2.0
Low temperature	2.25
Vibration	
Pointer oscillation	2.0
Pointer variation	1.5

Source: https://assist.dla.mil -- Downloaded: 202640-20T04:56Z
Check the source to verify that this is the current version before use.

ght Air Force Sustainment Center - Oklahoma City

3.6.1 Markings. The following marking shall be finished in color No. 37875 of FED-STD-595. The dimensions of the markings shall be as follows:

	Height or Length Inch ±0.016	Width of Line or Graduation Inch ± 0.005
Numerals 0,20,40,60 80 and 100	0.156	0.025
10-pound graduations	0.188	0.031
5-pound graduations	0.094	0.020
Lettering OIL PRESS	0.125	0.025
Lettering PSI	0.125	0.025
Mask reference lines	----	0.020
Pointer reference mark	0.063	0.015

3.6.1.2 Black markings. The markings USAF and TYPE MO-2 shall be permanently and legibly marked on the dial in letters 0.063 inch in height. Unless otherwise specified, these and all other markings shall be finished in color No. 37038 of FED-STD-595.

3.7 Electrical wiring. The internal and external wiring shall conform to figure 2 and shall be connected to the receptable pins as shown thereon. The internal wiring shall be color coded.

3.8 Weight. The weight of the indicator shall not exceed 0.50 pound.

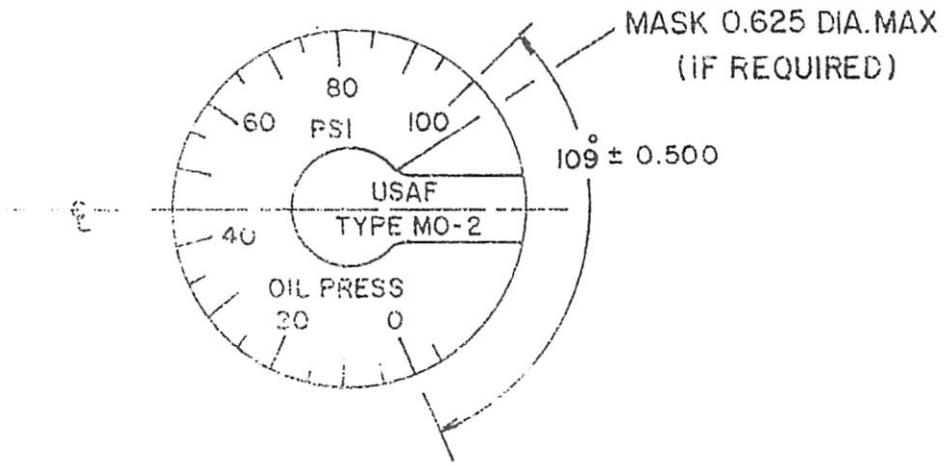
3.9 Synthetic bearings. When sapphire or ruby synthetic bearings are used, they shall conform to MIL-B-27497.

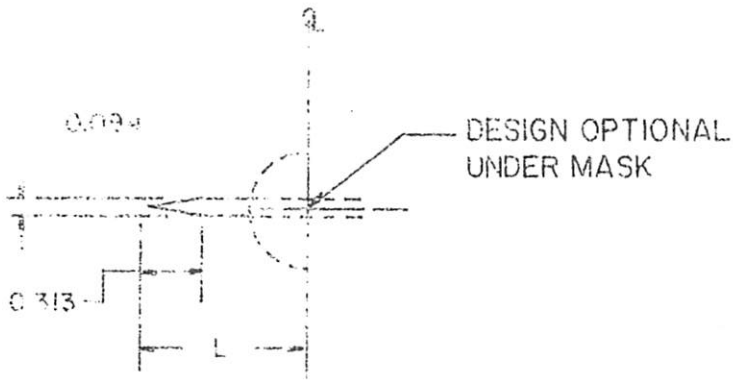
3.10 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-20T04:56Z
Check the source to verify that this is the current version before use.





(IF MASK NOT REQUIRED, USE MS33585-5 POINTER

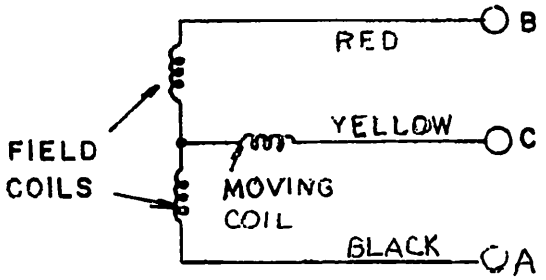
DIMENSIONS IN INCHES

UNLESS OTHERWISE SPECIFIED, TOLERANCES FRACTION ± 0.016

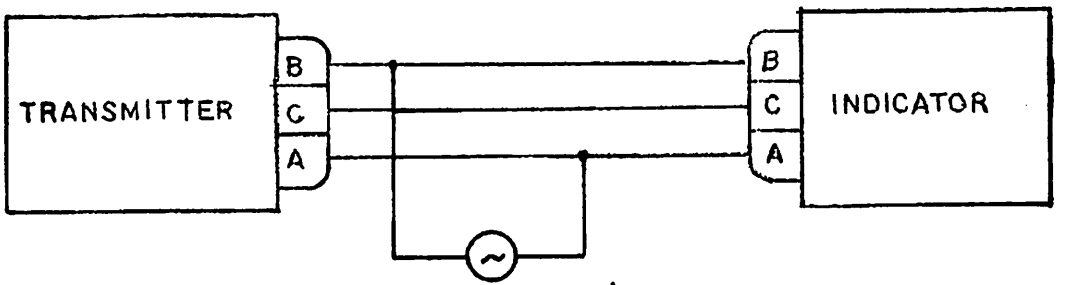
FIG. 5. Dual and pointer.

Source: https://assist.dla.mil - Downloaded: 20260-20T04:56Z
Check the source to verify that this is the current version before use.

Air Force Sustainment Center - Oklahoma City



INTERNAL WIRING



EXTERNAL WIRING

FIGURE 2. Wiring diagram.

Source: https://assist.dla.mil - Downloaded: 202610-20T04:56Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.1.1 General specification. The inspection and testing of indicators shall be in accordance with MIL-I-25437 and as specified herein. The indicator shall operate within the tolerance specified in 3.4.1.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.3.2)

4.3 Qualification inspection. Unless otherwise specified all inspections shall be performed in accordance with the test conditions specified in this specification.

4.3.1 Test samples. The qualification test samples shall consist of ten indicators representative of the production equipment. Three of the samples shall be subjected to all the tests except reliability and the other seven samples shall be subjected to the reliability test only. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

4.3.2 Qualification conformance inspections. The qualification conformence inspections shall consist of all the test specified herein and in MIL-I-25437.

4.4 Test methods

4.4.1 Reliability. Unless otherwise specified, the seven indicators specified in 4.3.1 shall be subjected to the following reliability test which shall demonstrate a minimum acceptable MTBF of 2,000 hours at a 90 percent confidence level. Acceptance permitting production shall be in accordance with MIL-STD-781, test plan I.

4.4.1.1 Test. The test equipment and fixtures shall include provisions for vibration and temperature cycling at the periodic intervals specified herein. During this test, inputs signals shall be applied to the indicator at a rate of 2 to 3 cycles (minimum and maximum) per minute from 0 to 85 percent of full scale and return. The test shall begin when rated power is applied. After completion of 4 hours of operation with rated power on, which shall include at least 480 cycles of operation, the power shall be turned off and shall remain off for a period of 10 minutes. For at least 10 minutes out of every hour of ON time, the indicator shall be vibrated in accordance with MIL-STD-781, test level J.

Source: https://assist.dla.mil -- Downloaded: 202610-20T04:56Z
Check the source to verify that this is the current version before use.

The temperature cycles shall be conducted in the following sequence: 4 hours at -20°C, 14 minutes for temperature change to +50°C at a minimum rate of 5°C per minute; 4 hours at +50°C, 14 minutes for temperature change to -20°C; and 4 hours at -20°C. In case of malfunction the malfunction shall be recorded and the test interrupted while repairs are made. Upon completion of each 250 hours of cycling, the indicator shall be checked for scale error. The scale error reading shall not exceed 150 percent of the room temperature tolerance in table I. No recalibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report in accordance with MIL-STD-785 which shall contain test procedures, details of any malfunctions, and corrective actions taken.

4.4.1.2 Longevity. At the completion of the reliability test, three of the indicators specified in 4.4.1 shall continue to be tested for a total of 3, hours each. The test time accumulated during the reliability test shall be applied to this test. The accept-reject criteria will not be applicable.

4.4.2 One indicator shall be submitted to the MIL-I-25438 specification preparing activity for a compatibility test. The indicator shall be connected to a variety of pressure transmitters which meet the requirements of MIL-T-38230. The indicator and transmitter shall be connected per Figure 2 of MIL-I-25438. The scale error test portion of MIL-T-25435 paragraph 4.5.3 at room temperature shall be performed at room temperature. The tolerance at each test point shall be the sum of the tolerances of MIL-I-25438 Table I and MIL-I-38230 Table I at room temperature.

4.5 Packaging of inspection. Tests of methods of packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2)

5.1.1 Level A

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T04:56Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5..3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type MO-2 indicators covered by this specification intended for use in conjunction with the TRU-66/A pressure transmitter in indicating remotely the pressure in oil systems of aircraft engines.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When sampling plan B will not be conducted,.

c. Levels of packaging and packing required (see section 5).

6.3 Qualification. With respect to products requiring qualifications, away will be made only for products which are, at the time set for opening of being qualified for inclusion in the applicable Qualified Products List whether not such products have actually been so listed by that date. The attention suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government for qualification in order that they may be eligible to be awarded contracting orders for the products covered by this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-20T04:56Z
Check the source to verify that this is the current version before use.

ght Air Force Sustainment Center - Oklahoma City

The activity responsible for the Qualified Products List is the OC-ALC/MMEDO, Tinker AFB OK 73145-5990 and information per to qualifications of products may be obtained from that activity.

6.4 Subject term (key word) listing.

Indicator
MIL-T-38230
MO2
Oil
Pressure
Reluctance
TRU 66A
0-100

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue order to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Preparing activity:
AIR FORCE -71

Project number:
6620-F441

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T04:56Z
Check the source to verify that this is the current version before use.